METRIC

FF-B-171/7

30 NOVEMBER 1993

FEDERAL SPECIFICATION SHEET

BEARING, BALL, ANNULAR, SINGLE ROW, RADIAL,

The complete requirements for acquiring the bearings described herein shall consist of this document and the latest issue of FF-B-171/GEN.

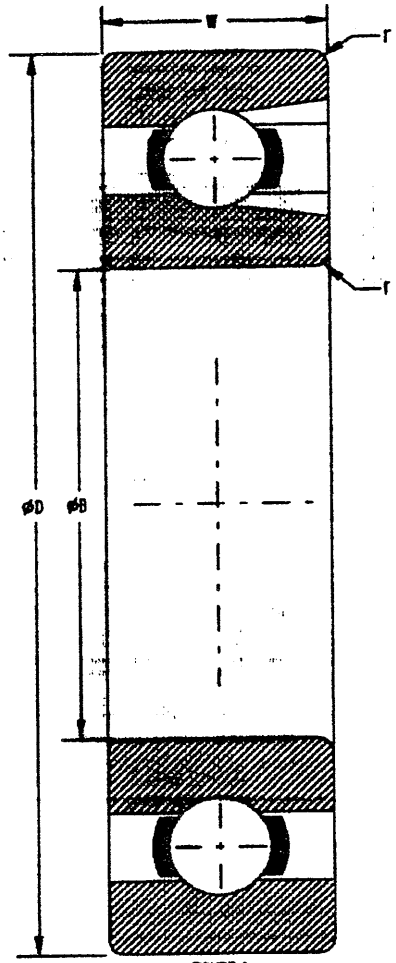


FIGURE 1.
OPEN (NO SHIELDS)
(CODE A)

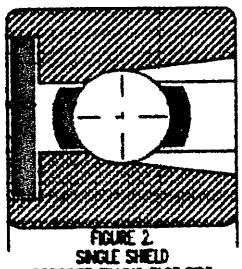


FIGURE 2.
SINGLE SHIELD
OPPOSITE FILLING SLOT SIDE
(NORMAL POSITION)
(CODE B)

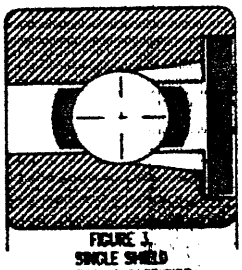


FIGURE 3.
SINGLE SHIELD
ON FILLING SLOT SIDE
(REVERSE POSITION)
(CODE C)

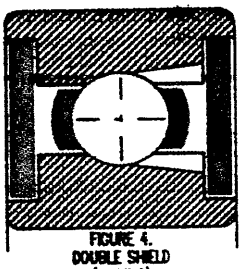


FIGURE 4.
DOUBLE SHIELD
(CODE D)

BEARING CONFIGURATIONS

AMSC/NA

FSC 3110

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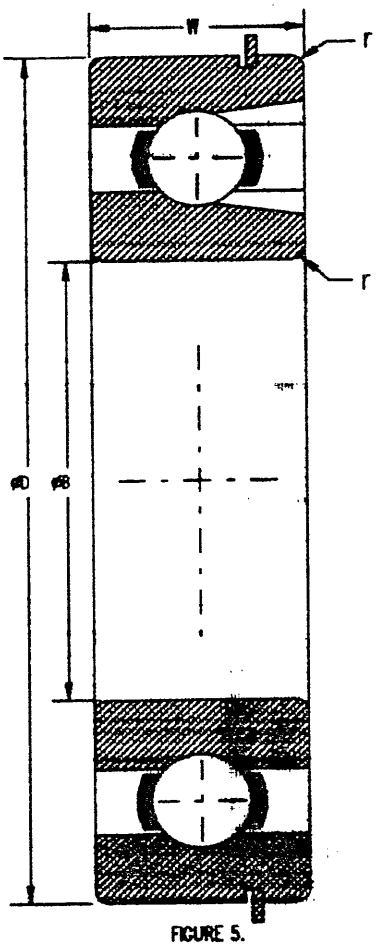


FIGURE 5.
OPEN WITH SNAP RING
ON FILLING SLOT SIDE
(NORMAL POSITION)
(CODE E)

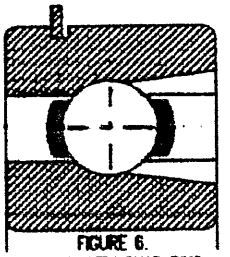


FIGURE 6.
OPEN WITH SNAP RING
OPPOSITE FILLING SLOT SIDE
(REVERSE POSITION)
(CODE F)

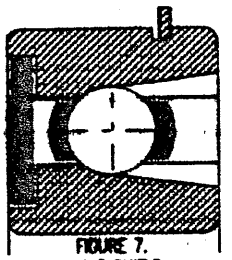


FIGURE 7.
SINGLE SHIELD,
NORMAL POSITION
WITH SNAP RING,
(NORMAL POSITION)
(CODE G)

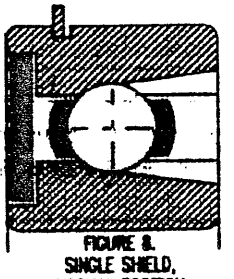


FIGURE 8.
SINGLE SHIELD,
NORMAL POSITION
WITH SNAP RING,
(REVERSE POSITION)
(CODE H)

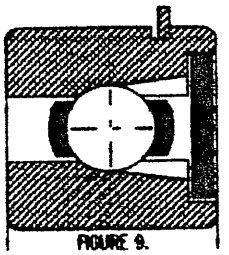


FIGURE 9.
SINGLE SHIELD,
REVERSE POSITION
WITH SNAP RING,
(NORMAL POSITION)
(CODE J)

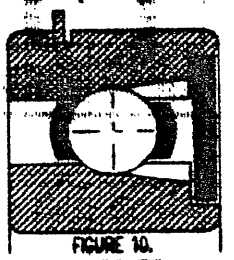


FIGURE 10.
SINGLE SHIELD,
REVERSE POSITION
WITH SNAP RING,
(REVERSE POSITION)
(CODE K)

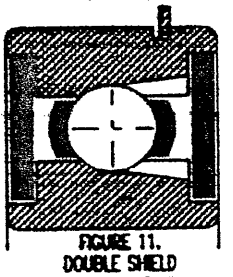


FIGURE 11.
DOUBLE SHIELD
WITH SNAP RING,
(NORMAL POSITION)
(CODE L)

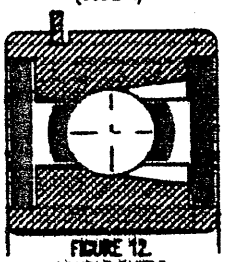


FIGURE 12.
DOUBLE SHIELD
WITH SNAP RING,
(REVERSE POSITION)
(CODE M)

BEARING CONFIGURATIONS - (Continued)

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REQUIREMENTS:

1. **Boundary dimensions.** The boundary dimensions shall be specified in the part number by the size codes in Table I.

2. Cages. Cages shall be manufactured from any material unless otherwise specified. Cages shall be specified in the part number by the codes in table II.

3. Shields and snap rings. Shields shall be fabricated from any material unless otherwise specified and shall be removable or nonremovable type. Snap rings shall be fabricated from ferrous material unless otherwise specified and shall have sufficient spring temper to set firmly in the ring grooves and shall be removable without the metal taking a permanent set. The snap ring and snap sing groove dimensions and their relative locations shall be in accordance with ANSI/AFBMA Standard 20. The number and positioning of shields and snap rings required shall be specified in the part number by the codes in table III.

4. Radial **Internal** Clearance. The tolerance limits for radial internal clearance shall be in accordance with ANSI/AFBMA Standard 20 and shall be specified in the part number by the codes in table IV.

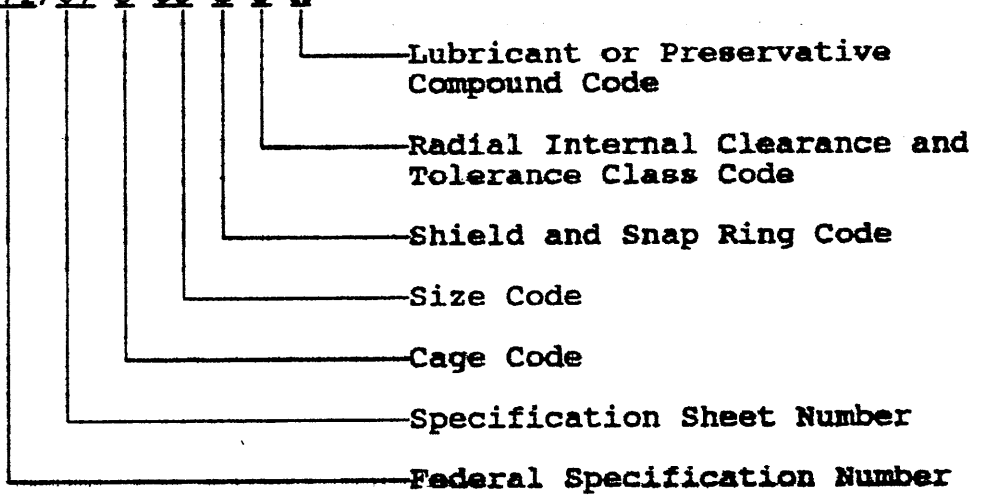
5. **Tolerance Class.** The tolerance limits for tolerance class ABEC-1 and ABEC-5 shall be in accordance with ANSI/AFBMA Standard 20 and shall be specified in the part number by the codes in table XV.

6. **Lubricant** or Preservative compound Preservative compound shall be applied to all open bearings (no shields or seals) and to single closure bearings (single shield) that have no additional lubricant requirement. Thus, for these bearings the bearing part number shall include the “C” suffix. For double closure or single closure bearings that require lubricant the bearing part number shall include the “A” or “B” suffix. When grease is required, the bearing void (the airspace between the inner and outer ring of the assembled bearing) shall be 25 to 50 percent filled. The lubricant or preservative compound shall be specified in the part number by the codes in table V.

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7. Part number. The part number consists of the letters FFB, the basic Federal specification number, the specification sheet number, and alpha or numeric codes for cage, size, shields and snap rings, radial internal clearance and tolerance class, and lubricant or preservative compound.

EXAMPLE: FFB171/07 J 05 C 5 A



FFB171/07JO5C5A indicates - Bearing, cage-pressd steel or molded plastic; bore 25mm; outside diameter 52mm; width 15mm; single shield on filling slot side (reverse position); radial internal clearance symbol 2 (less than normal); tolerance class ABEC-5; lubricant in accordance with MIL-G-81322.

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TABLE I. Boundary dimensions.

Size Code 1/	OB Bore	OD Outside Diameter	W Width	Chamfer $r_s$min 2/	Shaft Shoulder Diameter Min 3/	C Dynamic load rating (pounds)
02	15	35	11	0.6	20	1695
03	17	40	12	0.6	22	1955
04	20	47	14	1.0	26	2360
05	25	52	15	1.0	31	3270
06	30	62	16	1.0	36	4320
07	35	72	17	1.1	42	5700
08	40	80	18	1.1	47	6790
09	45	85	19	1.1	52	7630
10	50	90	20	1.1	57	7980
11	55	100	21	1.5	63.5	9870
12	60	110	22	1.5	68.5	11935
13	65	120	23	1.5	73.5	13020
14	70	125	24	1.5	78.5	14155
15	75	130	25	1.5	83.5	14735
16	80	140	26	2.0	90	17255
17	85	150	28	2.0	95	18540
18	90	160	30	2.0	100	22800
19	95	170	32	2.1	107	24740
20	100	180	34	2.1	112	27795
21	105	190	36	2.1	117	30100
22	110	200	38	2.1	122	31950
24	120	215	40	2.1	132	35300
26	130	230	40	3.0	144	38000

1/ Size codes in Table I are codes.

2/ The minimum bearing chamfer ($r_s$min) must clear the maximum shaft or housing filler radius. This dimension does not control the shape of the bearing chamfer contour.

3/ For reference purposes only. Maximum shoulder heights shall be such as to permit bearing removal pressure to be applied to the bearing inner ring. In cases where this requirement cannot be met, provisions shall be made for the removal of the bearing from the shaft without damage to the bearing or other parts.

TABLE II. Codes for Cages.

Code	Cage
J	Pressed steel or molded plastic
M	Machined bronze or brass
T	Machined non-metallic (phenolic)

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TABLE III. Codes for Shields and Seals.

Code	Shield and snap ring configuration
A	open (no shields) - fig.1
B	single shield opposite filling slot side, (normal position) - fig. 2
C	single shield on filling slot side, (reverse position) - fig. 3
D	double shield - fig. 4
E	open with snap ring on filling slot side, (normal position) - fig. 5
F	open with snap ring opposite filling slot side, (reverse position) - fig. 6
G	single shield, normal position, with snap ring, (normal position) - fig. 7
H	single shield, normal position, with snap ring, (reverse position) - fig. 8
J	single shield, reverse position, with snap ring, (normal position) - fig. 9
K	single shield, reverse position, with snap ring, (reverse position) - figure 10
L	double shield with snap ring, (normal position)-fig. 11
M	double shield with snap ring, (reverse position)-fig. 12

TABLE IV. Codes for Radial Internal Clearance and ABEC tolerance.

Code	Radial Internal Clearance	Tolerance Class
1	Symbol 2 (less than normal)	ABEC-1
2	Symbol 0 (normal)	
3	Symbol 3 (greater than normal)	
4	Symbol 4 (greater than symbol 3)	
5	Symbol 2 (less than normal)	ABEC-5
6	Symbol 0 (normal)	
7	Symbol 3 (greater than normal)	
8	Symbol 4 (greater than symbol 3)	

TABLE V. Codes for Lubricant and Preservative Compound.

Code	Lubricant or Preservative Compound
A	Grease in accordance with MIL-G-81322
B	Grease in accordance with DOD-G-24508
C	Preservative compound in accordance with MIL-P-197

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